Customer Shopping Behavior Analysis: Project Report

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Abstract

This project analyses 3,900 retail purchases worth $233,081 to understand who buys what, how they pay and ship, and which offers matter. The data was cleaned and enriched in Python (pandas), stored in a MySQL database, and presented in a one-page Power BI dashboard. The main finding is that spending is remarkably even: average order value stays between $57 and $62 across every category, age group, gender, season and subscription status. Differences lie in volume instead. Clothing and Accessories bring 76.5% of revenue, men account for 67.7% of sales, and only 26.9% of revenue comes from subscribers.

Introduction and objectives

Retailers collect large amounts of purchase data but rarely turn it into decisions. This project takes one customer shopping dataset and answers practical business questions with it.

Objectives:

- Clean and prepare the raw data so it is reliable for analysis.
- Build a MySQL database so the data can be queried with SQL.
- Find patterns in spending, product popularity, ratings, subscriptions and discounts.
- Present the results in an interactive Power BI dashboard.
- Turn the findings into recommendations a retailer could act on.

Dataset description

The dataset has 3,900 rows, one per customer purchase, and 18 original columns with no duplicate customers (Customer ID runs 1 to 3,900). Only Review Rating had missing values (37 rows).

Column	Type	Meaning
Customer ID	Integer	Unique customer number
Age	Integer	18 to 70 years, mean 44.1
Gender	Text	Male (2,652) or Female (1,248)
Item Purchased	Text	25 products, such as Blouse, Jeans, Sandals
Category	Text	Clothing, Accessories, Footwear, Outerwear
Purchase Amount (USD)	Integer	$20 to $100, mean $59.76
Location	Text	50 US states
Size, Color	Text	4 sizes and 25 colors
Season	Text	Spring, Summer, Fall, Winter
Review Rating	Decimal	2.5 to 5.0, mean 3.75
Subscription Status	Text	Yes or No
Shipping Type	Text	6 options, such as Express and Free Shipping
Discount Applied, Promo Code Used	Text	Yes or No
Previous Purchases	Integer	1 to 50 earlier orders
Payment Method	Text	6 options, such as PayPal and Credit Card
Frequency of Purchases	Text	7 values, from Weekly to Annually

Tools and technologies

Tool	Used for
Python 3.13	Main programming language
pandas	Loading, cleaning and transforming the data

Tool	Used for
SQLAlchemy and PyMySQL	Connecting Python to MySQL and writing the table
MySQL	Storing the cleaned data and running SQL queries
Jupyter Notebook (VS Code)	Writing and documenting the analysis
Power BI Desktop	Building the interactive dashboard

Data cleaning and feature engineering

Five steps turned the raw CSV into an analysis-ready table of 3,900 rows and 19 columns.

1. **Inspected the data** with `head()`, `info()`, `describe()` and `isnull().sum()`. This showed 37 missing Review Ratings and no other gaps.
2. **Filled missing ratings** with the median rating of each product category, which avoids letting one category's ratings distort another's.
3. **Standardised column names** to lowercase with underscores, and renamed `Purchase Amount (USD)` to `purchase_amount`, so they work cleanly in SQL.
4. **Created age groups** with `pd.qcut` into four equal-sized quartiles: Young Adult (18 to 31), Adult (32 to 44), Middle Aged (45 to 57) and Senior (58 to 70).
5. **Converted purchase frequency to days** (Weekly = 7, Fortnightly and Bi-Weekly = 14, Monthly = 30, Quarterly and Every 3 Months = 90, Annually = 365) to create a numeric `frequency_of_days` column.

The notebook also found that `discount_applied` and `promo_code_used` match in all 3,900 rows. The duplicate `promo_code_used` column was dropped.

MySQL integration

The cleaned DataFrame was loaded into a MySQL database named `customer_behavior`, in a table called `customer`. Python created the database if needed (`CREATE DATABASE IF NOT EXISTS`), connected through a SQLAlchemy engine using the PyMySQL driver, and wrote the table with `df.to_sql()`. A `SELECT * FROM customer LIMIT 5` query confirmed the load.

Example query used for product ratings:

```
SELECT item_purchased,
       ROUND(AVG(review_rating), 2) AS `Average Product Rating`
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

Power BI then connected to the same database through the MySQL connector.

Analysis and key findings

The dataset holds 3,900 orders worth $233,081, with an average order of $59.76 and an average rating of 3.75 out of 5.

Revenue by category

Clothing and Accessories together bring 76.5% of revenue. Order values are almost identical, so volume drives the gap.

Category	Orders	Revenue (USD)	Share	Avg order (USD)	Avg rating
Clothing	1,737	104,264	44.7%	60.03	3.72
Accessories	1,240	74,200	31.8%	59.84	3.77
Footwear	599	36,093	15.5%	60.26	3.79
Outerwear	324	18,524	7.9%	57.17	3.75

Revenue by age group

The youngest group (18 to 31) is the largest and has the highest average rating, but the spread across groups is small, from 23.9% to 26.7%.

Age group	Orders	Revenue (USD)	Share	Avg order (USD)	Avg rating
Young Adult (18 to 31)	1,028	62,143	26.7%	60.45	3.80
Middle Aged (45 to 57)	986	59,197	25.4%	60.04	3.71
Adult (32 to 44)	942	55,978	24.0%	59.42	3.73
Senior (58 to 70)	944	55,763	23.9%	59.07	3.75

Gender, subscription and discounts

Men make 67.7% of purchases. Subscribers are only 27% of customers and spend and rate the same as non-subscribers. Discounts were used on 42.7% of orders but did not lift order value or ratings.

Segment	Orders	Revenue share	Avg order (USD)	Avg rating
Male	2,652	67.7%	59.54	3.75
Female	1,248	32.3%	60.25	3.74
Subscribed	1,053	26.9%	59.49	3.75
Not subscribed	2,847	73.1%	59.87	3.75
Discount used	1,677	42.7%	59.28	3.74
No discount	2,223	57.3%	60.13	3.76

Season, products and locations

- **Season:** Fall earns the most ($60,018, 25.7%) and has the highest average order ($61.56). Summer is the lowest ($55,777, 23.9%).
- **Top products by revenue:** Blouse ($10,410), Shirt ($10,332), Dress ($10,320), Pants ($10,090) and Jewelry ($10,010).
- **Highest rated products:** Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80) and Handbag (3.78).
- **Top states by revenue:** Montana ($5,784), Illinois ($5,617), California ($5,605), Idaho ($5,587) and Nevada ($5,514).
- **Shipping and payment:** all six shipping types and all six payment methods each hold between 15.7% and 17.5% of revenue. Free Shipping and Credit Card lead by a small margin.

Power BI dashboard

The dashboard is a single 1920 by 1080 page that lets a manager filter the whole business view in a few clicks. It reads the `customer` table from MySQL.

- **KPI cards:** Average Purchase Amount, Average Review Rating and Number of Customers.

- **Slicers (filters):** Subscription Status, Gender, Category and Shipping Type.
- **Subscription donut:** percentage of customers by subscription status.
- **Category charts:** Revenue by Category and Sales by Category (customer count), as column charts.
- **Age group charts:** Revenue by Age Group and Sales by Age Group (customer count), as bar charts.

Every visual responds to the slicers, so a user can, for example, select Female customers who are subscribed and see how revenue by category changes.

Business recommendations

1. **Grow Outerwear and Footwear.** They bring only 7.9% and 15.5% of revenue. Footwear has the best average rating (3.79) among categories, so promoting it carries little quality risk.
2. **Attract more women.** Women make 32.3% of purchases but spend as much per order as men ($60.25 against $59.54). Targeted campaigns could lift volume without changing pricing.
3. **Rethink subscriptions.** Only 26.9% of revenue comes from subscribers, and they spend the same as others. Adding real member benefits could give people a reason to subscribe and buy more.
4. **Use discounts more selectively.** Discounted orders were slightly smaller ($59.28 against $60.13) and rated the same, so across-the-board discounts may be giving away margin.
5. **Plan for seasons.** Fall has the highest average order ($61.56) and Summer the lowest revenue. Stock and promotions can follow this pattern.
6. **Feature the best-rated items.** Gloves, Sandals and Boots lead on rating and are good candidates for homepage and email placement.

Conclusion and future work

The project shows that this customer base behaves very evenly: no segment spends meaningfully more per order, so growth has to come from more orders, a broader category mix and a better-targeted use of discounts and subscriptions. The pipeline from CSV to Python to MySQL to Power BI is reusable for any similar retail dataset.

Limitations: the data has no purchase dates, so trends over time cannot be measured, and the differences between segments are small and were not tested for statistical significance.

Future work:

- Add significance tests (for example, a t-test on discounted against full-price orders).
- Segment customers with clustering (such as RFM or k-means) using previous purchases and frequency.
- Build a model to predict subscription status or high-value purchases.
- Add order dates to enable seasonality and trend analysis.